SIMATS ENGINEERING
ASSESSMENT TOOL 1

COURSE NAME: COMPUTER NETWORKS
COURSE CODE: CSA07 14

COURSE OUTCOME COVERED

CO2: Configure IP addressing schemes, subnetting, and basic routing techniques using simulation tool, for efficient network design and topology. *(BL3)*

Assessment Tool Weightage: 40%

QUESTIONS: 5 **Total Marks:50**

Annexure A

ANALYTICAL PROBLEM SOLVING

Code of Conduct:

I Sandhiya S (192521224) (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.
I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:

Annexure A

ANALYTICAL PROBLEM SOLVING

1. A university network is assigned the IP address block 192.168.20.0/24 for its internal communication. The network administrator has decided to divide this network into subnets for three different labs using fixed subnet masks. Lab A is assigned a /26 subnet, Lab B is assigned a /27 subnet, and Lab C is assigned a /28 subnet. Based on the given subnet masks, apply subnetting concepts to determine the network address of each lab. Further, identify the valid range of host IP addresses and the broadcast address for each subnet. Finally, calculate the total number of usable host IP addresses available in each lab's subnet.

2. A network administrator has configured two systems with the following details: System 1 has IP address 192.168.1.130 with subnet mask 255.255.255.192, and System 2 has IP address 192.168.1.190 with subnet mask 255.255.255.192. Using the given configuration, apply subnetting techniques to determine the subnet range, network address, and broadcast address for each system. Identify the valid host IP range for both subnets based on the given subnet mask. Calculate whether each IP address falls within its respective subnet range. Also, determine the number of usable host addresses available in each subnet.

3. An organization has been allocated the IP block **172.16.0.0/16** for its internal network. The network administrator has already decided the subnet masks for each department as follows: HR → /25, Sales → /26, IT → /27, and Admin → /28. Using the given subnet masks, apply subnetting techniques to assign suitable subnet addresses for each department. Determine the network address, valid host IP range, and broadcast address for each subnet. Based on the allocation, calculate the number of usable host addresses in each department.

Finally, identify the remaining unused IP address range within the given network block.

4. A company uses the IP address **172.16.10.0/24** and has implemented subnetting using a **/27 subnet mask** for its internal departments. Using the given subnet mask, apply subnetting techniques to calculate the total number of subnets created and the number of usable hosts per subnet. Determine the subnet to which the IP address **172.16.10.78** belongs. Identify the network address and broadcast address for that subnet. Also, determine whether the IP address **172.16.10.95** falls within the same subnet range based on the given configuration.

5. A network uses the Internet Protocol version 6 address **2001:0db8:0000:0000:0000:ff00:0042:8329**. Apply IPv6 addressing rules to compress the given address into its shortest form and then expand the compressed address back to its full notation. Using a prefix length of **/64**, determine the network prefix of the given address. Also, calculate the total number of possible host addresses available within this network based on the given prefix length.

RUBRICS FOR CALCULATION

Criteria	Weightage	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)
Problem Understanding	20%	Clearly interprets problem, identifies all parameters and requirements accurately	Understands problem with minor gaps	Partial understanding; misses key elements	Misinterprets or unable to understand problem
Method Selection	20%	Selects most appropriate method/formula with strong justification	Selects correct method with minor errors	Inappropriate or partially correct method	Unable to select method
Solution Process	25%	Logical, systematic steps with correct approach throughout	Mostly correct steps with minor mistakes	Disorganized steps; multiple errors	No clear process
Accuracy of Calculation	20%	All calculations accurate; correct final answer	Minor calculation errors	Frequent errors; incorrect final answer	No valid calculations
Interpretation of Results	15%	Clearly interprets results with units and engineering relevance	Basic interpretation with minor omissions	Limited or unclear interpretation	No interpretation

Computer Network [CSA0714]

Name : Sandhiya S
Reg No : 192521224

Analytical Problem

Given
A university network has been assigned to the IP address block.

192.168.20.0/24.

The network is divided among three labs.

Lab A -> /26
Lab B -> /27
Lab C -> /28.

Step 1

The network 192.168.20.0/24 has

Subnet mask = 255.255.255.0

Total addresses = 2^{32-24}

$= 2^8$

Total address = 256.

$\therefore$ Network address = 192.168.20.0
Broadcast address = 192.168.20.265
Total IPs = 256.

The administrator now divides this block into smaller subnets.

Step 2

/26 means:

255.255.255.192.

Number of Addresses :

$= 2^{32-26}$

$= 2^6$

$= 64$

Total addresses = 64

64 - 2 = 62.

Block size

256 - 192 = 64

Subnet increments

0, 64, 128, 192.

Lab A Details

Network Address = 192.168.20.0/26.

Result

Item	Value
Network Address	192.168.20.0
First host	192.168.20.1
Last host	192.168.20.62
Broadcast Address	192.168.20.63
Usable Hosts	62.

Step 3.

The next available address after Lab A is

192.168.20.64.

Subnet mask

/27

255.255.255.224

$= 2^{32-27}$

$= 2^5$

$= 32$

Total addresses = 32

usable hosts

32 - 2 = 30.

Block size

256 - 224 = 32.

Result.

Item	Value.
Network Address	192.168.20.64
First host	192.168.20.65
Last host	192.168.20.94
Broadcast Address	192.168.20.95
usable hosts	30.

Step - 4

The next available address after Lab B is

192.168.20.96.

Number of Addresses.

$= 2^{32-28}$

= 16.

usable host

16 - 2 = 14

Block size

256 - 240 = 16.

Result

Item	Value
Network Address	192.168.20.96
First host	192.168.20.97
Last host	192.168.20.110
Broadcast Address	192.168.20.111
usable host	14.

Step 5

Lab	CIDR	Subnet mask	Network Addresses	Host Range	Broad cast Address	usable host.
Lab A	/26	255.255.255.192	192.168.20.0	192.168.20.1 - 192.168.20.62	192.168.20.63	62
Lab B	/27	255.255.255.224	192.168.20.64	192.168.20.65 192.168.20.94	192.168.20.65	30
Lab C	/28	255.255.255.240	192.168.20.96	192.168.20.97 192.168.20.110	192.168.20.111	14.

Total usable host addresses allocated:

62 + 30 + 14 = 106

∴ The three labs together can support 106 usable host devices.

Subnetting the Network

172.16.0.0/16.

Given
An Organization has been allocated the network.

Department	Subnet.
HR	/25
Sales	/26
IT	/27
Admin	/28

Department	Network Address	Valid Host Range	Broadcast Address	Usable Host.
HR (\25)	172.16.0.0	172.16.0.1 - 172.16.0.126	172.16.0.127	126.
Sales (\26)	172.16.0.128	172.16.0.129 - 172.16.0.190	172.16.0.191	62
IT (\27)	172.16.0.192	172.16.0.193- 192.16.0.222	172.16.0.223	30
Admin (\28)	172.16.0.224	172.16.0.225 - 172.16.0.238	172.16.0.239	14.

Calculation

HR (\25) : $2^7 - 2 = 126$

Sales (\26 : $2^6 - 2 = 62$

IT(\27) : $2^5 - 2 = 30$
Admin (\28) : $2^4 - 2 = 14$.

The network 172.16.0.0\16 is successfully subnetted into the four departmental networks.
The departments receive 126, 62, 30 and 14 usable host addresses respectively, giving a total 232 usable host address across all department.